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HOMEWORK #1.

Problem 1. (LSIRA 1.1.2)

Assume that the sum of 2 integers x and y is even. Show that x and y are either both even or both odd.

Proof.

For contrapositive proof, assume that

$A =$ "sum of 2 integers x and y is even"

$B =$ "x and y have the same property (pattern) in terms of being odd or even (both even or both odd)"

We have to show if A, then B, that is

$A \Rightarrow B$ (1).

Instead of proving (1), we prove not $A \Leftarrow$ not B, that is

$\neg B =$ not $B = \{$ x and y do not have the same property $\}$
(one is odd and one is even)

$\neg A =$ not $A = \{$ sum of x and y is not even $\}$
(sum of x & y is odd)

So, we need to show $\neg B \Rightarrow \neg A$. (2)

Assume either x or y is odd and the other is even. Order does not matter. Let's say x is odd and y is even $\Rightarrow$

$x = 2k+1$ for some nonnegative integer k. (3)

$y = 2m$ for some nonnegative integer m. (4)

$x+y = 2k+1+2m = 2(k+m)+1$ (5)

$(x+y) = 2(k+m)+1$ which is odd. $\Rightarrow$

Similar case happens if we say x is even and y is odd. $\Rightarrow$

$\neg B = \neg A$ (6) $\Rightarrow$

$A \Rightarrow B$ (7)

So, if the sum of $(x+y)$ is even, for x and y there is no way of being one odd and the other even.

Problem 2. (LSIRA 1.1. Prob. 4)

Prove basic properties of rational numbers. Recall that a real number r is <u>rational</u> if $r = \frac{a}{b}$ where a, b are integers and $b \neq 0$. A real number that is not rational is called <u>irrational</u>.

(a) Show that if r, s are rational numbers, so are $(r+s)$, $(r-s)$, rs and $\frac{r}{s}$ (provided $s \neq 0$)

(b) Assume r is rational and a is irrational. Show that $(r+a)$ and $(r-a)$ are irrational. Show also that if $r \neq 0$, then ra, $\frac{r}{a}$, $\frac{a}{r}$ are irrational.

(c) Show by example that if a, b are irrational numbers, then $(a+b)$ and (ab) can be rational or irrational depending on a & b.

Proof. (Using direct proof)

(a) Assume r & s are rational numbers $\Rightarrow$

$r = \frac{a}{b}$ (1) where a, b are integers and $b \neq 0$ (2)

$s = \frac{c}{d}$ (3) where c, d are integers and $d \neq 0$ (4)

$$(r \pm s) = \frac{a}{b}^{\backslash d} \pm \frac{c}{d}^{\backslash b} = \frac{\overbrace{ad}^{\in \mathbb{Z}} \pm \overbrace{bc}^{\in \mathbb{Z}}}{\underbrace{bd}_{\in \mathbb{Z}}} = \frac{m}{n} \in \mathbb{Q} \quad \begin{matrix} m, n \in \mathbb{Z} \\ n \neq 0 \end{matrix}$$

recall $bd \neq 0$, $bd \in \mathbb{Z}$ considering (2) and (4).

$$r \cdot s = \frac{a}{b} \cdot \frac{c}{d} = \frac{\overbrace{ac}^{\in \mathbb{Z}}}{\underbrace{bd}_{\in \mathbb{Z}}} = \frac{p}{q} \in \mathbb{Q}$$

Let's see $\frac{r}{s}$ provided $s \neq 0 \Rightarrow$

$$s = \frac{c}{d} \neq 0 \Rightarrow c \neq 0 \quad (5)$$

Then,

$$\frac{r}{s} = \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{\overbrace{a \cdot d}^{\in \mathbb{Z}}}{\underbrace{bc}_{\in \mathbb{Z}}} = \frac{k}{l} \in \mathbb{Q} \quad \begin{matrix} k, l \in \mathbb{Z} \\ l \neq 0 \end{matrix}$$

Again, $l = bc \neq 0$ considering (2) and (5).

Also, $\frac{s}{r} = \frac{1}{\frac{r}{s}} = \frac{\frac{c}{d}}{\frac{a}{b}} = \frac{bc}{a \cdot d} = \frac{l}{k} \in \mathbb{Q}$ provided $(r \neq 0)$. (6)

(b) Assume r is rational $\Rightarrow$

$$r = \frac{d}{b} \quad (1) \quad \text{where } d, b \in \mathbb{Z} \text{ and } b \neq 0. \quad (2)$$

Also assume

a is irrational. (3)

(b.1) Show that $(r \pm a)$ irrational.

Proof:

Considering (1), (2), (3), assume that

$(r \pm a)$ is rational. $\Rightarrow$

$(r \pm a) = \frac{m}{n}$ (4) for some integers $m, n \in \mathbb{Z}$, $n \neq 0$.

Say $\frac{m}{n} = s$ (s - rational) (5) $\Rightarrow$

$r \pm a = s \Rightarrow a = (s \mp r)$ (6) for

rational s and r.

Using the result from part (a), for rational numbers s & r $(s \pm r)$ is rational, that is

$a = (s \mp r)$ is rational (7)

which is contradiction. a is irrational.

So $(r \pm a)$ are irrational.

(b.2) Show also that if $r \neq 0$, then ra, $\frac{r}{a}$ and $\frac{a}{r}$ are irrational.

Proof.

For contradiction, assume ra is rational under (1)-(3) assumptions and $r \neq 0$ (8). $\Rightarrow$

$ra = s$ (9) where $s = \frac{m}{n} \in Q$ $\Rightarrow$

$\frac{ra}{r} = \frac{s}{r}$ (10) we can divide both sides by r, since $r \neq 0$ based on (8).

$a = \frac{s}{r}$ (11) for some rational s, r

Using the result from part (a), for rational s, r $\frac{s}{r} \in Q$ is a rational number which means

$a = \frac{s}{r} \in Q \Rightarrow a$ is rational number too,

which is contradiction, a is irrational. $\Rightarrow$

$(r \cdot a)$ is irrational. (12)

Similarly,

$\frac{r}{a}$ & $\frac{a}{r}$ are irrational numbers too.

We show $\frac{r}{a}$ is irrational from which we can conclude $\frac{a}{r}$ is also irrational since $\frac{a}{r} = \frac{1}{\left(\frac{r}{a}\right)} = \frac{1}{k}$

k is irrational. Dividing by irrational k does not change its irrationality.

For contradiction, suppose $\frac{r}{a}$ is rational, under (1) - (3) assumptions and $r \neq 0$ (8)

$\frac{r}{a} = s$ (13) where $s \in Q$, and $s \neq 0$ (14) as $r \neq 0$

$\frac{r}{a} = s \Rightarrow a = \frac{r}{s}$ (15) for $s \in Q$, $s \neq 0$.

$a = \frac{r}{s}$ (15) for some rational r, s.

Using the result from part (a), for rational numbers r, s, $\frac{r}{s} \in Q$ is a rational number which means

$a = \frac{r}{s} \in Q$ (15) $\Rightarrow$ a is a rational number too,

which is a contradiction, a is irrational $\Rightarrow$

$\left(\frac{r}{a}\right)$ is irrational. (16)

We can say $\left(\frac{a}{r}\right)$ is irrational too based on the argument we made in the beginning. Or irrationality of $\left(\frac{a}{r}\right)$ can be shown in the similar way above.

(c) Show by example that if a, b are irrational numbers, then $(a+b)$ and (ab) can be rational or irrational, depending on a & b.

Proof.

To have rational outcomes:

$$\begin{cases} a = (3-\sqrt{2}) \in \mathbb{I} \\ b = (3+\sqrt{2}) \in \mathbb{I} \end{cases} \quad (1)$$

$\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$

($\mathbb{I}$ - set of irrational numbers)

$$a+b = (3-\sqrt{2}) + (3+\sqrt{2}) = 6 \in \mathbb{Q} \quad (2)$$

$$a \cdot b = (3-\sqrt{2}) \cdot (3+\sqrt{2}) = 7 \in \mathbb{Q} \quad (3)$$

To have irrational outcomes

$$\begin{cases} a = \sqrt{3} \in \mathbb{I} \\ b = \sqrt{2} \in \mathbb{I} \end{cases} \quad (4) \Rightarrow \quad \begin{aligned} (a+b) &= (\sqrt{3}+\sqrt{2}) \in \mathbb{I} \quad (5) \\ a \cdot b &= \sqrt{3} \cdot \sqrt{2} = \sqrt{6} \in \mathbb{I} \quad (6) \end{aligned}$$

$\square$

Problem 4.

Prove (1.2.2) formula:

(Distributive laws). For all sets $B, A_1, A_2 \ldots A_n$

$$B \cup \left(\bigcap_{i=1}^{n} A_i\right) = \bigcap_{i=1}^{n} (B \cup A_i) \qquad (1)$$

Proof.

To prove (1), using the subset inclusion method, we need to show

(a) $B \cup \left(\bigcap_{i=1}^{n} A_i\right) \subseteq \bigcap_{i=1}^{n} (B \cup A_i) \qquad (2)$ and

(b) $B \cup \left(\bigcap_{i=1}^{n} A_i\right) \supseteq \bigcap_{i=1}^{n} (B \cup A_i) \qquad (3)$

First,

(a) ($\subseteq$)

Take $\forall x \in B \cup \left(\bigcap_{i=1}^{n} A_i\right) \qquad (4) \Rightarrow$

$x \in B$ or $x \in \bigcap_{i=1}^{n} A_i$ (using def-n of $\cup$) $\Rightarrow$

$\underbrace{x \in B \text{ or } x \in A_i}_{\Downarrow}$, for all A_i (using def-n of $\cap$) $\Rightarrow$

$x \in B \cup A_i$ for every A_i (using def-n of $\cup$) $\Rightarrow$

$\Rightarrow x \in \bigcap_{i=1}^{n} (B \cup A_i)$ which is what we wanted to prove

$\forall x \in B \cup (\bigcap_{i=1}^{n} A_i) \Rightarrow x \in \bigcap_{i=1}^{n} (B \cup A_i) \Rightarrow$

$B \cup (\bigcap_{i=1}^{n} A_i) \subseteq \bigcap_{i=1}^{n} (B \cup A_i)$ (def-n of a subset) (5)

Second,

(b) (2)

Take $\forall x \in \bigcap_{i=1}^{n} (B \cup A_i)$ (6) $\Rightarrow$

$x \in B \cup A_i$ for every A_i (def-n of $\cap$) $\Rightarrow$

$x \in B$ or $x \in A_i$ for every A_i (def-n of $\cup$) $\Rightarrow$

$x \in B$ or $x \in \bigcap_{i=1}^{n} A_i$ (def-n of $\cap$) $\Rightarrow$

$x \in B \cup (\bigcap_{i=1}^{n} A_i)$ (def-n of $\cup$) which is what we wanted to prove

$\forall x \in \bigcap_{i=1}^{n} (B \cup A_i) \Rightarrow x \in B \cup (\bigcap_{i=1}^{n} A_i) \Rightarrow$

$\bigcap_{i=1}^{n} (B \cup A_i) \subseteq B \cup (\bigcap_{i=1}^{n} A_i)$ (def-n of a subset) (6)

$\square$